

Code Explanation

Configuration Settings for ETL Pipeline

The provided code defines configuration settings for an ETL (Extract, Transform, Load) pipeline using Python dataclasses. It includes two main classes: `DataSource` and `PipelineConfig`, which are used to store configuration data for data sources and the ETL pipeline, respectively.

Overview of the Code

This code is designed to manage the configuration of an ETL pipeline. It provides a structured way to define data sources and pipeline settings, making it easier to manage and modify the pipeline's configuration.

Step 1: Defining the DataSource Class

The `DataSource` class is defined to represent a data source in the ETL pipeline. It includes attributes for the data source's name, path, format, and options.

```
@dataclass
class DataSource:
    """Configuration for a data source."""

    name: str
    path: str
    format: str
    options: Dict[str, str] = None

    def __post_init__(self):
        if self.options is None:
            self.options = {}
```

Step 2: Initializing DataSource Options

The `\_\_post\_init\_\_` method is used to initialize the `options` attribute to an empty dictionary if it is not provided during the creation of a `DataSource` object.

```
def __post_init__(self):
    if self.options is None:
        self.options = {}
```

Step 3: Defining the PipelineConfig Class

The `PipelineConfig` class is defined to represent the configuration of the ETL pipeline. It includes attributes for the data sources, target path, target format, checkpoint path, write mode, and partition columns.

```
@dataclass
class PipelineConfig:
    """Configuration for the ETL pipeline."""

    sources: List[DataSource]
    target_path: str
```

```
target_format: str = "delta"
checkpoint_path: Optional[str] = None
write_mode: str = "overwrite"
partition_by: Optional[List[str]] = None
```

Step 4: Creating a Default Pipeline Configuration

A default pipeline configuration is created using the `PipelineConfig` class, which includes three data sources and other pipeline settings.

```
DEFAULT_CONFIG = PipelineConfig(
    sources=[
        DataSource(
            name="sales",
            path="dbfs:/mnt/data/raw/sales/",
            format="csv",
            options={"header": "true", "inferSchema": "true"}
        ),
        DataSource(
            name="products",
            path="dbfs:/mnt/data/raw/products/",
            format="parquet"
        ),
        DataSource(
            name="customers",
            path="dbfs:/mnt/data/raw/customers/",
            format="delta"
        )
    ],
    target_path="dbfs:/mnt/data/processed/sales_analytics/",
    target_format="delta",
    checkpoint_path="dbfs:/mnt/data/checkpoints/sales_analytics/",
    partition_by=["date", "region"]
)
```